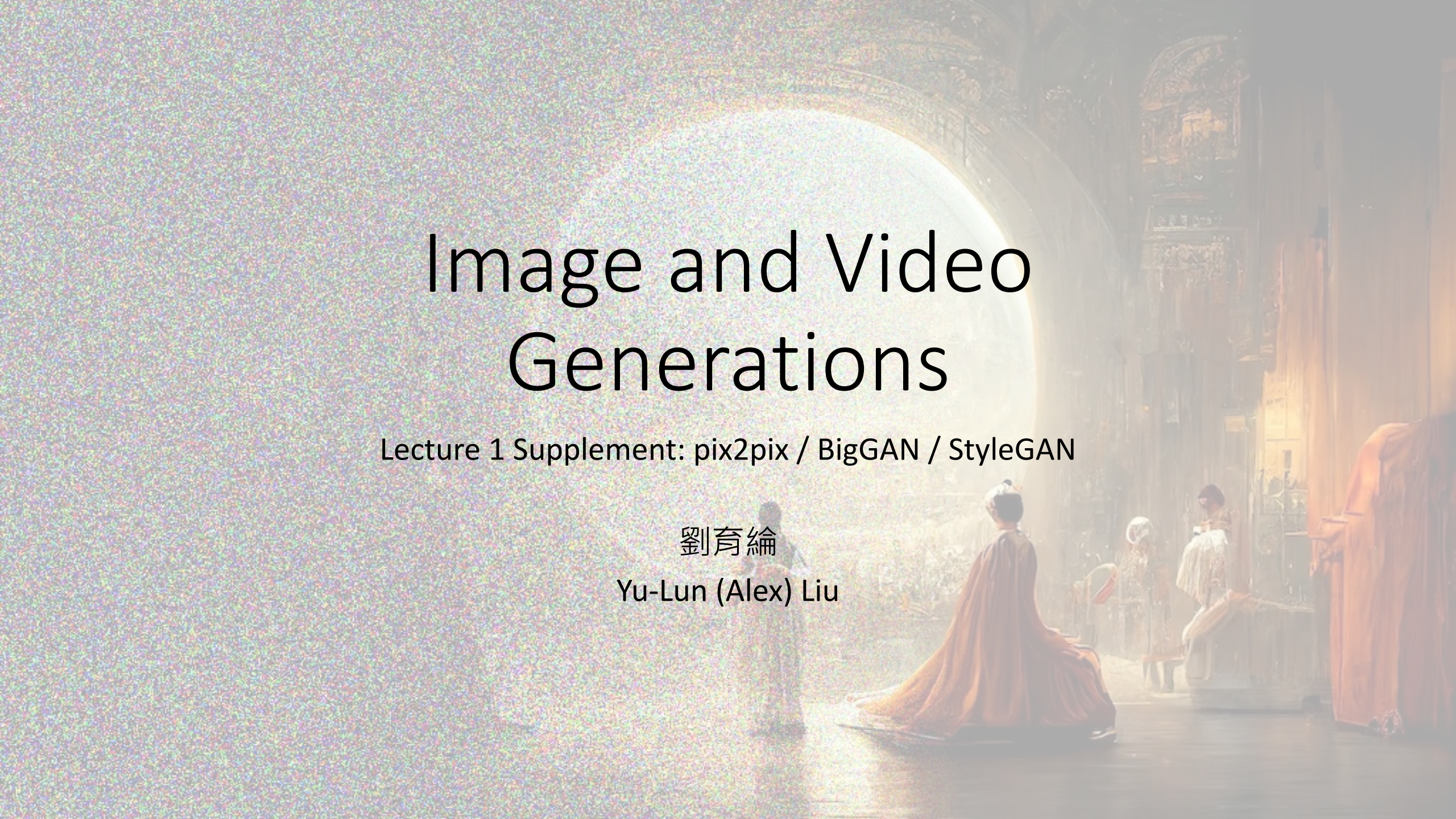


Image and Video Generations

Lecture 1 Supplement: pix2pix / BigGAN / StyleGAN

劉育綸

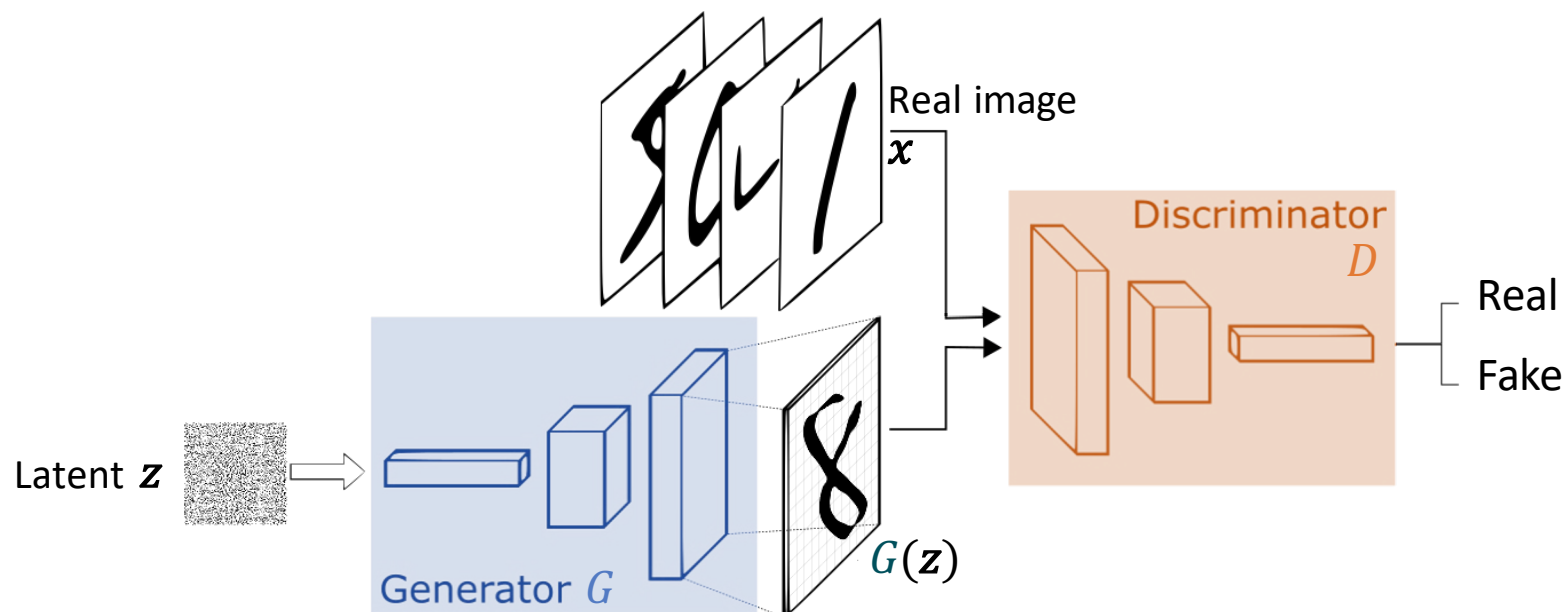
Yu-Lun (Alex) Liu



Recap: GAN

Loss function:

$$\min_G \max_D V(D, G) = \mathbb{E}_{x \sim p(x)} [\log D(x)] + \mathbb{E}_{z \sim p(z)} [\log(1 - D(G(z)))]$$



Recap: GAN

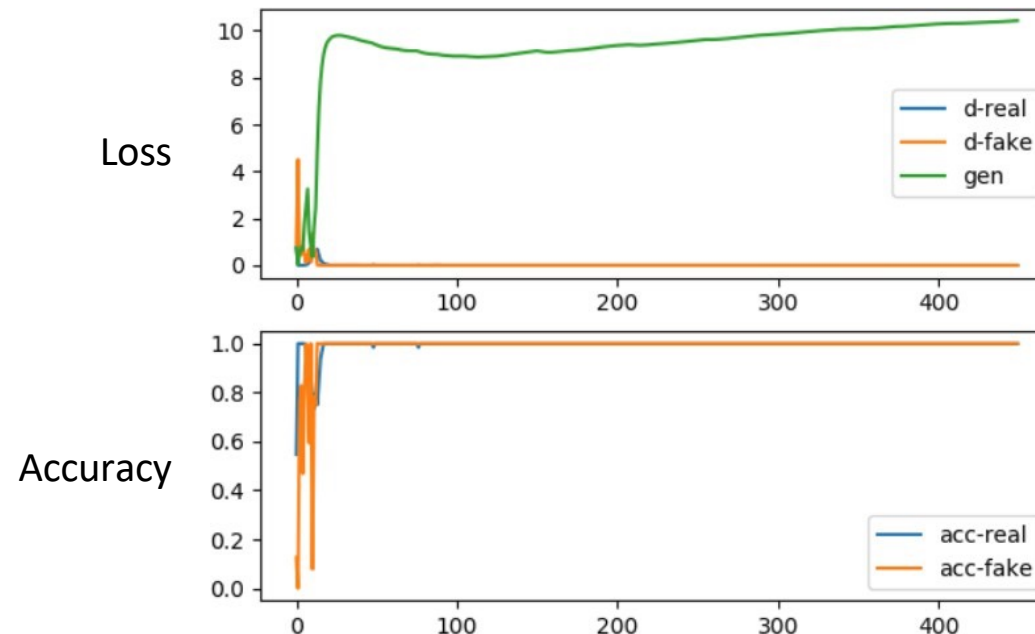
Loss function:

$$\min_{\underline{G}} \max_D V(D, G) = \mathbb{E}_{x \sim p(x)} [\log D(x)] + \mathbb{E}_{z \sim p(z)} [\log(1 - D(G(z)))]$$

A min-max (minimax) optimization problem, which is known as very difficult to solve!

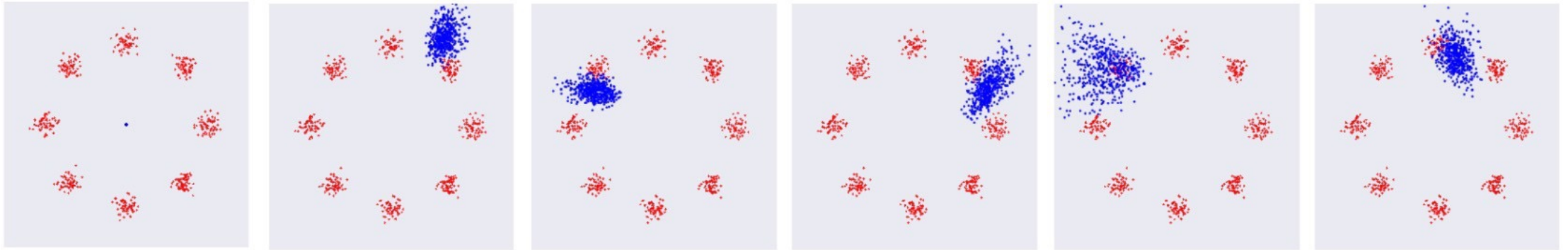
Recap: Non-Convergence

The **discriminator becomes too strong** and can easily distinguish between real and fake samples, leading to near- zero gradients for the generator and halting its learning.



Recap: Mode Collapse

The generator tends to **collapses** to a few modes of the data distribution instead of capturing the full diversity of the true data distribution.



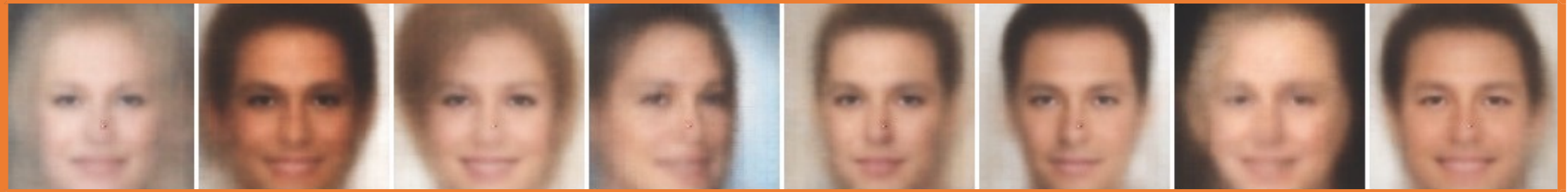
Recap: Limitations of VAEs

Typical failure cases of VAEs

Real
images



Generated
Images



Recap: StyleGAN2



Timeline

- [2014.06] GAN
- [2016.11] **pix2pix** (CVPR 2017)
 - **Conditional image-to-image translation**
- [2018.09] **BigGAN** (ICLR 2019)
 - **Scaling**
- [2018.12] **StyleGAN** (CVPR 2019)
 - **Controlling GAN**
- [2019.12] StyleGAN2 (CVPR 2020)

pix2pix

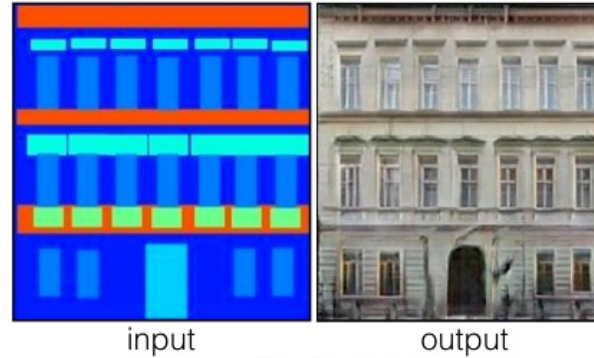
Isola et al., "Image-to-Image Translation with Conditional Adversarial Networks",
CVPR 2017.

Image-to-Image Translation

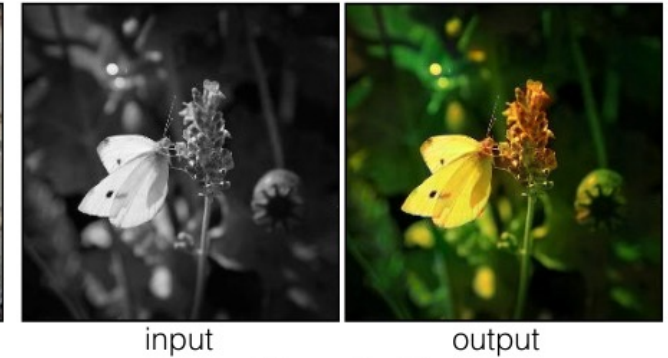
Labels to Street Scene



Labels to Facade



BW to Color



Aerial to Map



Day to Night



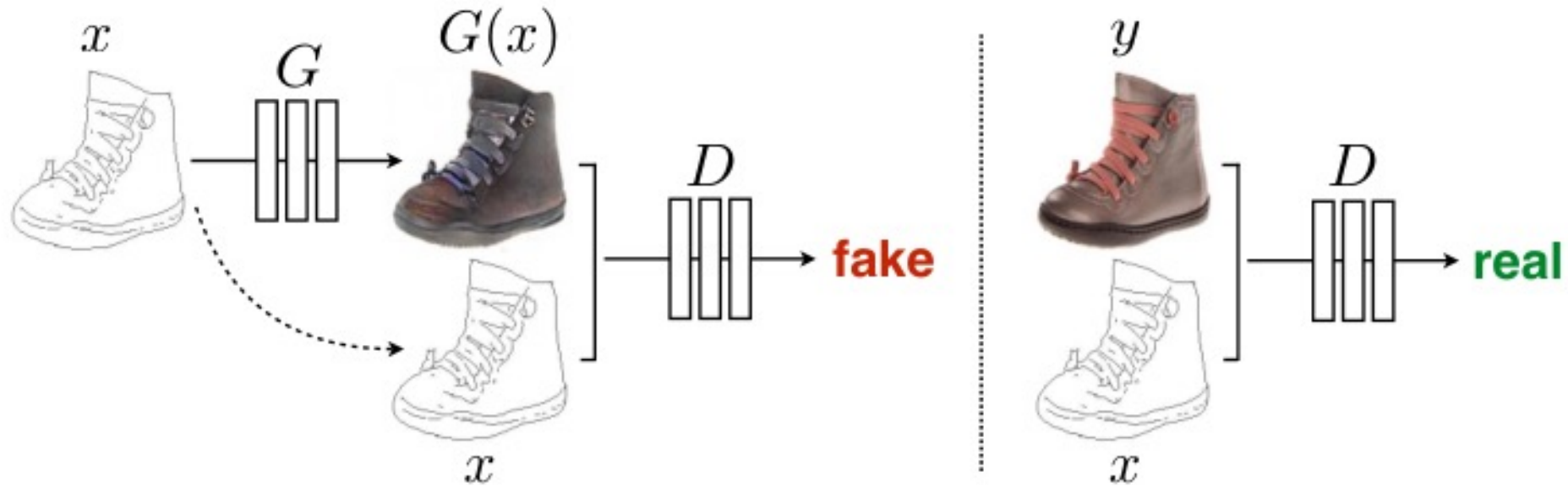
Edges to Photo



Related Work



Method: Conditional GAN

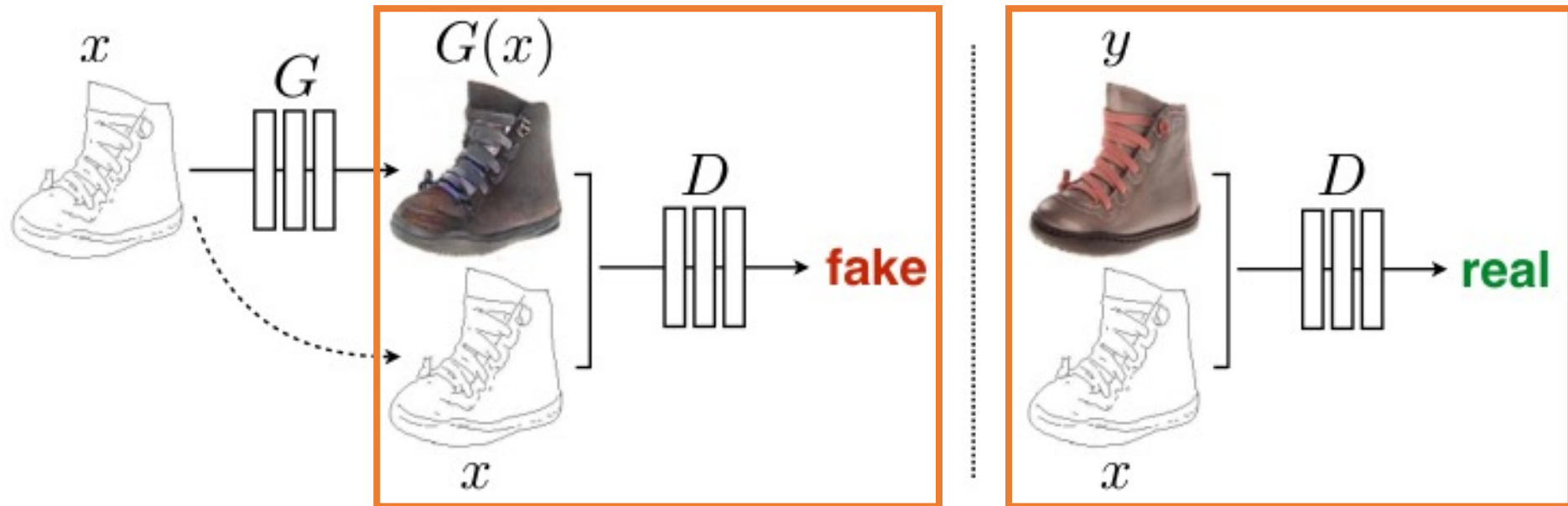


$$\mathcal{L}_{cGAN}(G, D) = \mathbb{E}_{x, y}[\log D(x, y)] + \mathbb{E}_{x, z}[\log(1 - D(x, G(x, z)))]$$

$$G^* = \arg \min_G \max_D \mathcal{L}_{cGAN}(G, D) + \lambda \mathcal{L}_{L1}(G)$$

$$\mathcal{L}_{L1}(G) = \mathbb{E}_{x, y, z}[\|y - G(x, z)\|_1], \quad \lambda = 100$$

Method: Conditional GAN



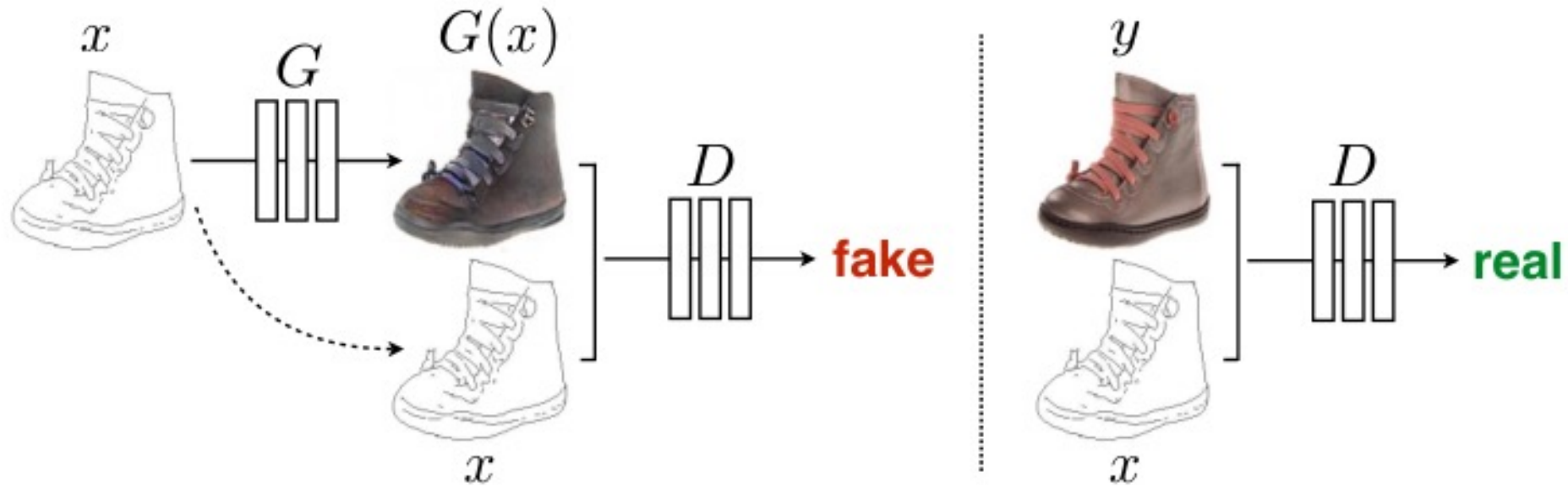
$$\mathcal{L}_{cGAN}(G, D) = \mathbb{E}_{x, y}[\log \underline{D(x, y)}] + \mathbb{E}_{x, z}[\log(1 - \underline{D(x, G(x, z))})]$$

$$G^* = \arg \min_G \max_D \mathcal{L}_{cGAN}(G, D) + \lambda \mathcal{L}_{L1}(G)$$

$$\mathcal{L}_{L1}(G) = \mathbb{E}_{x, y, z}[\|y - G(x, z)\|_1], \quad \lambda = 100$$

D 同時看 $\{x, y\}$ 並判斷：
「這個輸出 y 是不是輸入 x 的真實對應？」

Method: Conditional GAN



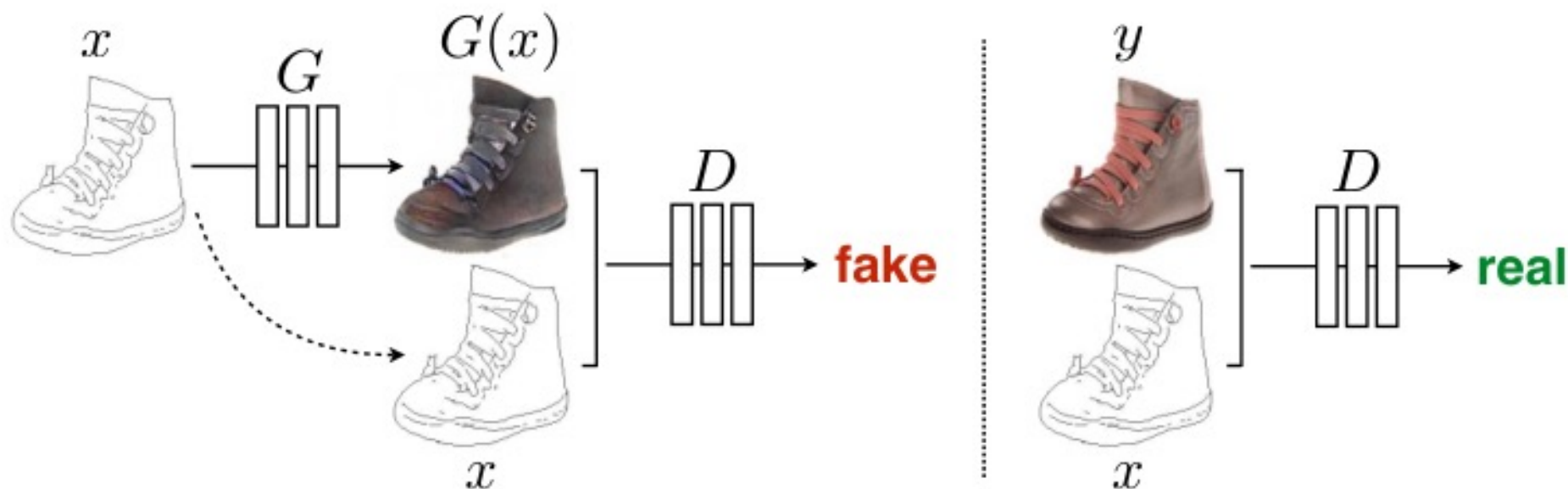
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$$\mathcal{L}_{L1}(G) = \mathbb{E}_{x, y, z}[\|y - G(x, z)\|_1], \quad \lambda = 100$$

GAN 項負責高頻銳利度
L1 負責低頻正確性

Method: Conditional GAN



$$\mathcal{L}_{cGAN}(G, D) = \mathbb{E}_{x, y}[\log D(x, y)] + \mathbb{E}_{x, z}[\log(1 - D(x, G(x, z)))]$$

$$G^* = \arg \min_G \max_D \mathcal{L}_{cGAN}(G, D) + \lambda \mathcal{L}_{L1}(G)$$

$$\mathcal{L}_{L1}(G) = \mathbb{E}_{x, y, z}[\|y - G(x, z)\|_1], \quad \lambda = 100$$

因為 L1 會鼓勵輸出 mode，只要有一點偏移（diversity）就會被 L1 懲罰，因此最後 G 學會忽略 z ，生成結果幾乎是 deterministic 的

Method: Conditional GAN

G 其實也真的不吃 z 了，Diversity 改成在 training 與 generation 時用 random dropout 做到



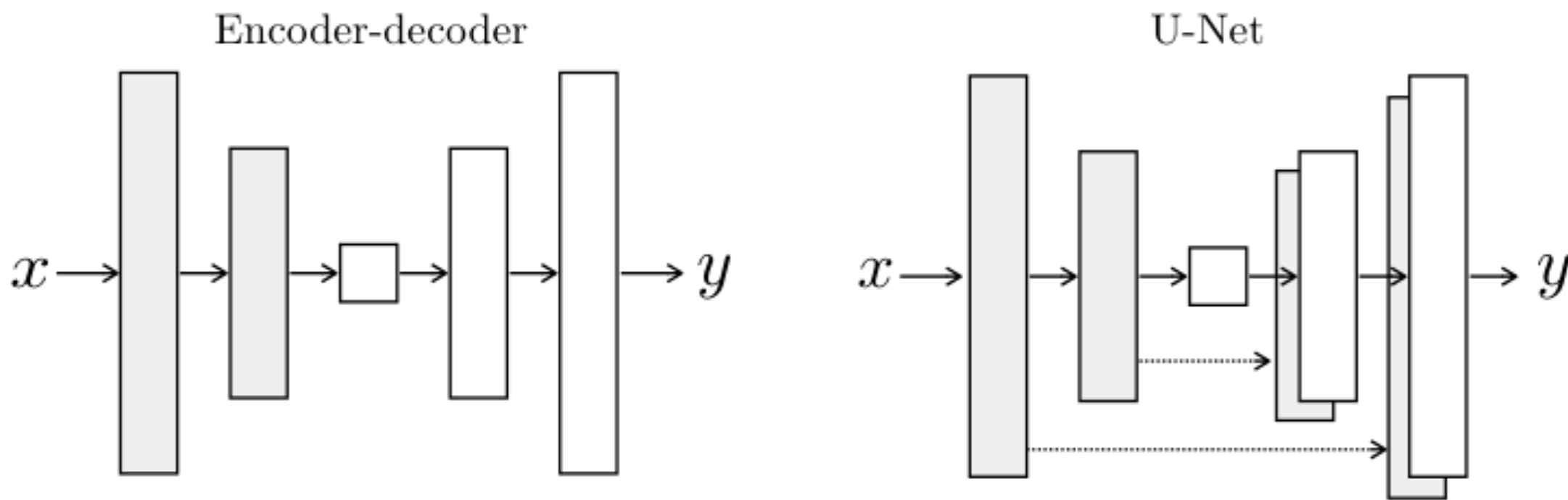
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$$\mathcal{L}_{L1}(G) = \mathbb{E}_{x, y, z}[\|y - G(x, z)\|_1], \quad \lambda = 100$$

Method:

U-Net Generator + PatchGAN discriminator



讓 input 與 output 的 low-level structure 也就是影像邊界對齊
skip connections 讓 low-level 資訊 bypass bottleneck

fully convolutional，所以模型與影像解析度無關 (train: 256x256，test: 512x512)

Method:

U-Net Generator + PatchGAN discriminator

因為 L1 已經顧好低頻了，所以 D 只需要管 high-frequency details
D 只看 $N \times N$ patch (預設 70×70)，convolutional 掃過整張圖取平均



相當於把 D 當成一種 texture / style loss 來使用增強細節

Ablation on Loss

Loss	Per-pixel acc.	Per-class acc.	Class IOU
L1	0.42	0.15	0.11
GAN	0.22	0.05	0.01
cGAN	0.57	0.22	0.16
L1+GAN	0.64	0.20	0.15
L1+cGAN	0.66	0.23	0.17
Ground truth	0.80	0.26	0.21

Table 1: FCN-scores for different losses, evaluated on Cityscapes labels $\leftrightarrow$ photos.

Ablation on Loss

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Table 1: FCN-scores for different losses, evaluated on Cityscapes labels $\leftrightarrow$ photos.

用 pre-trained 的 segmentation network 能否從生成影像 segment 出正確的 label

Ablation on the Loss Function

沒有 conditioning 的 GAN 會 collapse，永遠都輸出同一張圖

Loss	Per-pixel acc.	Per-class acc.	Class IOU
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Table 1: FCN-scores for different losses, evaluated on Cityscapes labels $\leftrightarrow$ photos.

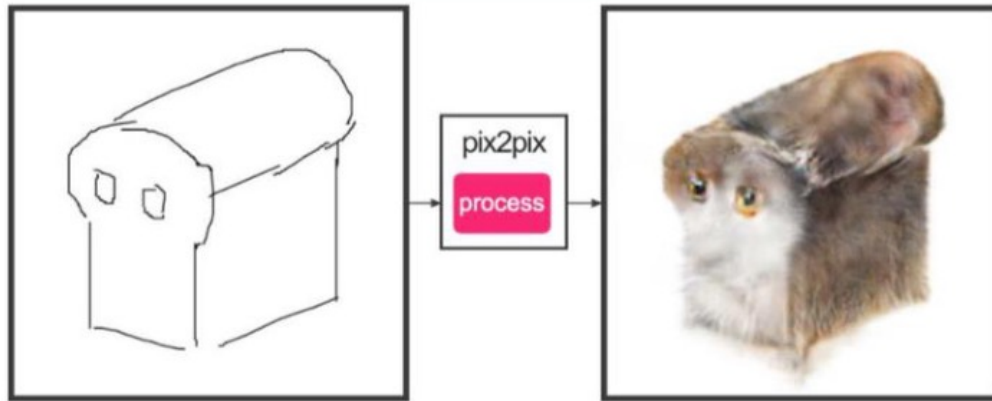
Ablation on the Discriminator Architecture

	Discriminator receptive field	Per-pixel acc.	Per-class acc.	Class IOU
(PixelGAN)	1×1	0.39	0.15	0.10
	16×16	0.65	0.21	0.17
(PatchGAN)	70×70	0.66	0.23	0.17
(ImageGAN)	286×286	0.42	0.16	0.11

Table 3: FCN-scores for different receptive field sizes of the discriminator, evaluated on Cityscapes labels→photos. Note that input images are 256×256 pixels and larger receptive fields are padded with zeros.

Example Applications by Online Community

#edges2cats by Christopher Hesse



sketch by Ivy Tsai

Background removal



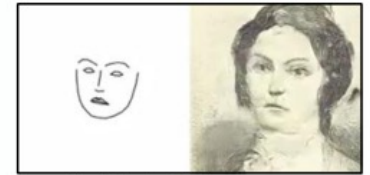
by Kaihu Chen

Palette generation



by Jack Qiao

Sketch → Portrait



by Mario Klingemann

Sketch → Pokemon



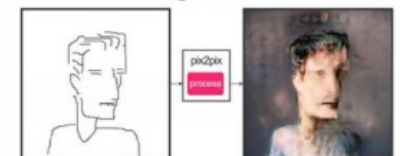
by Bertrand Gondouin

“Do as I do”



by Brannon Dorsey

#fotogenerator



sketch by Yann LeCun

PatchGAN 的後續

CycleGAN、pix2pixHD，甚至 Stable Diffusion 的 autoencoder 都是用 patch-based adversarial loss

Limitations

- 需要 paired data
 - CycleGAN (2017) 用 cycle consistency 做 unpaired translation
- 輸出 deterministic，z 被忽略
 - BicycleGAN (2017)，還有 StyleGAN 之後給每層都注入 noise 來增加細節與隨機性
- 解析度 256x256
 - pix2pixHD (2018) 做到 2048x1024

pix2pix 有輸入影像作為結構 condition

如果只有 ImageNet 1000 種 class label 作為 condition，可以用 GAN 生成高品質又 diverse 的影像嗎？

BigGAN

Brock et al., "Large Scale GAN Training for High Fidelity Natural Image Synthesis", ICLR 2019.

Class-Conditional Generation

從給定的 ImageNet 1000 種類別，生成 high fidelity 且多樣的影像



Related Work



(a)

容易的類別

(b)

困難的類別

Method: Scaling

Batch	Ch.	Param (M)	Shared	Skip- z	Ortho.	Itr $\times 10^3$	FID	IS
256	64	81.5	SA-GAN Baseline			1000	18.65	52.52
512	64	81.5	✗	✗	✗	1000	15.30	58.77(± 1.18)
1024	64	81.5	✗	✗	✗	1000	14.88	63.03(± 1.42)
2048	64	81.5	✗	✗	✗	732	12.39	76.85(± 3.83)
2048	96	173.5	✗	✗	✗	295(± 18)	9.54(± 0.62)	92.98(± 4.27)
2048	96	160.6	✓	✗	✗	185(± 11)	9.18(± 0.13)	94.94(± 1.32)
2048	96	158.3	✓	✓	✗	152(± 7)	8.73(± 0.45)	98.76(± 2.84)
2048	96	158.3	✓	✓	✓	165(± 13)	8.51(± 0.32)	99.31(± 2.10)
2048	64	71.3	✓	✓	✓	371(± 7)	10.48(± 0.10)	86.90(± 0.61)

Table 1: Fréchet Inception Distance (FID, lower is better) and Inception Score (IS, higher is better) for ablations of our proposed modifications. *Batch* is batch size, *Param* is total number of parameters, *Ch.* is the channel multiplier representing the number of units in each layer, *Shared* is using shared embeddings, *Skip- z* is using skip connections from the latent to multiple layers, *Ortho.* is Orthogonal Regularization, and *Itr* indicates if the setting is stable to 10^6 iterations, or it collapses at the given iteration. Other than rows 1-4, results are computed across 8 random initializations.

Method: Scaling

Inception Score:
只衡量像不像某一個類別的影像，不管多樣性

Batch	Ch.	Param (M)	Shared	Skip- z	Ortho.	Itr $\times 10^3$	FID	IS
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Batch size x8: IS +46% , 每個 batch 涵蓋更多 mode , 但 training 更早 collapse

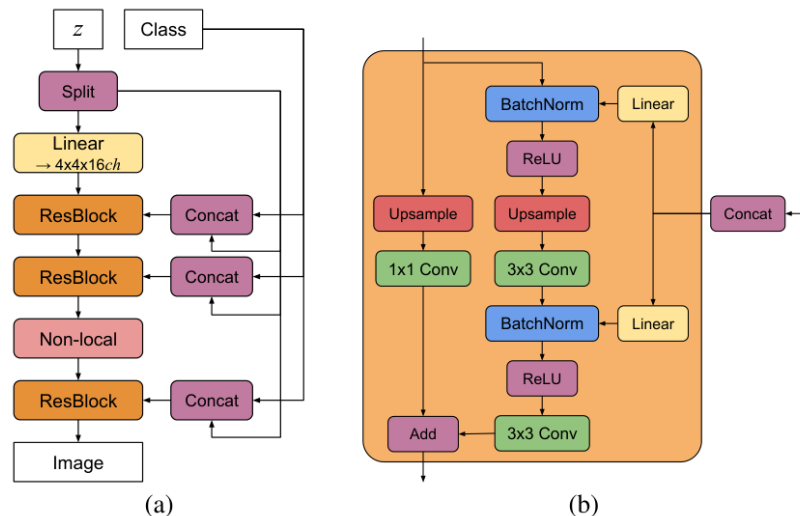
Method: Scaling

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2048	64	71.3	✓	✓	✓	371(± 7)	10.48(± 0.10)	86.90(± 0.61)

Network channel 數加寬 x1.5: IS +21% , 加深沒有幫助

Method: Scaling

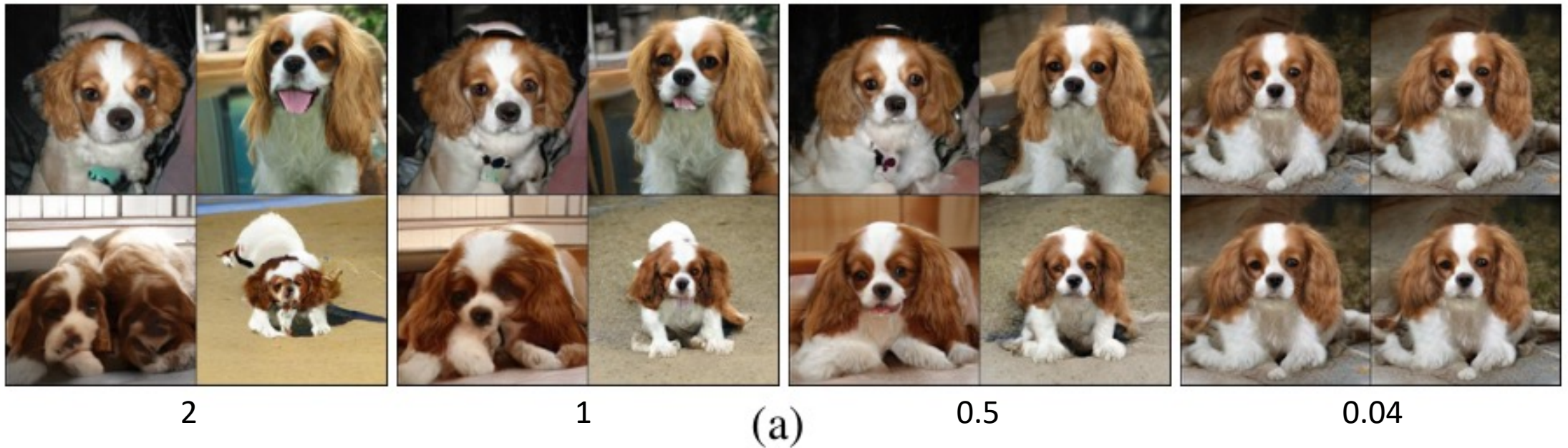
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Condition class embedding c 與 z 切塊從每一層輸入，輸入的方式是線性投影成該層 BatchNorm 的 gain 與 bias (Feature-wise Linear Modulation)，可以加速訓練收斂

這種用每層的 normalization scale/shift 來注入 condition 的方式很通用，在 StyleGAN 與 diffusion 也會看到

Method: Truncation Trick

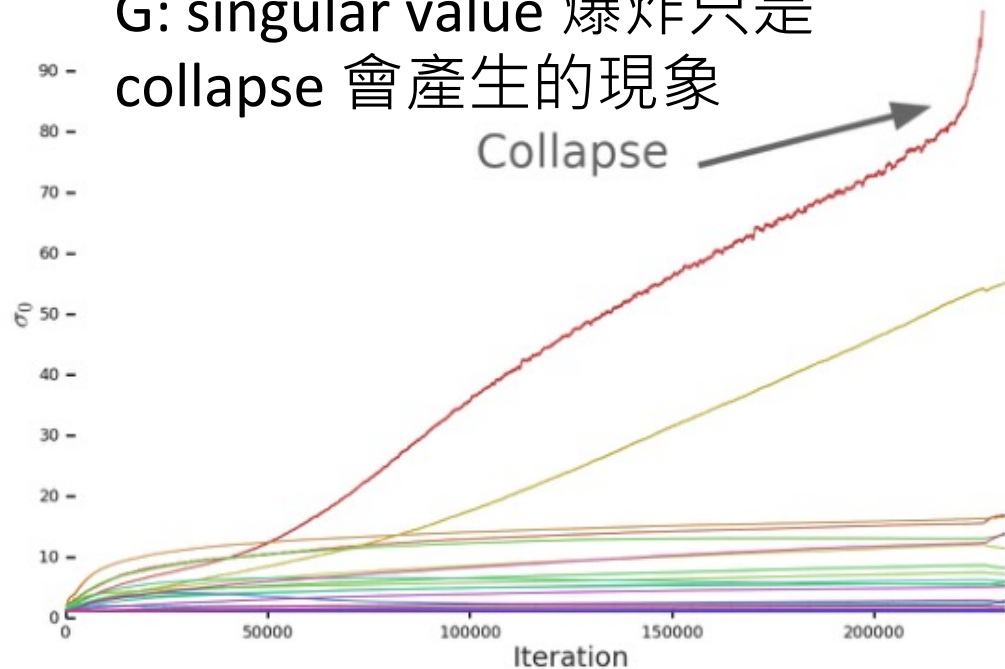


Training 時的 z 用 standard normal，但 generation 時改用 truncated normal

效果是 generate 的 sample 會更接近 mode，因此 IS 會上升，但 diversity 下降
(所以可以在 generation 時 trade-off between fidelity 與 variety)

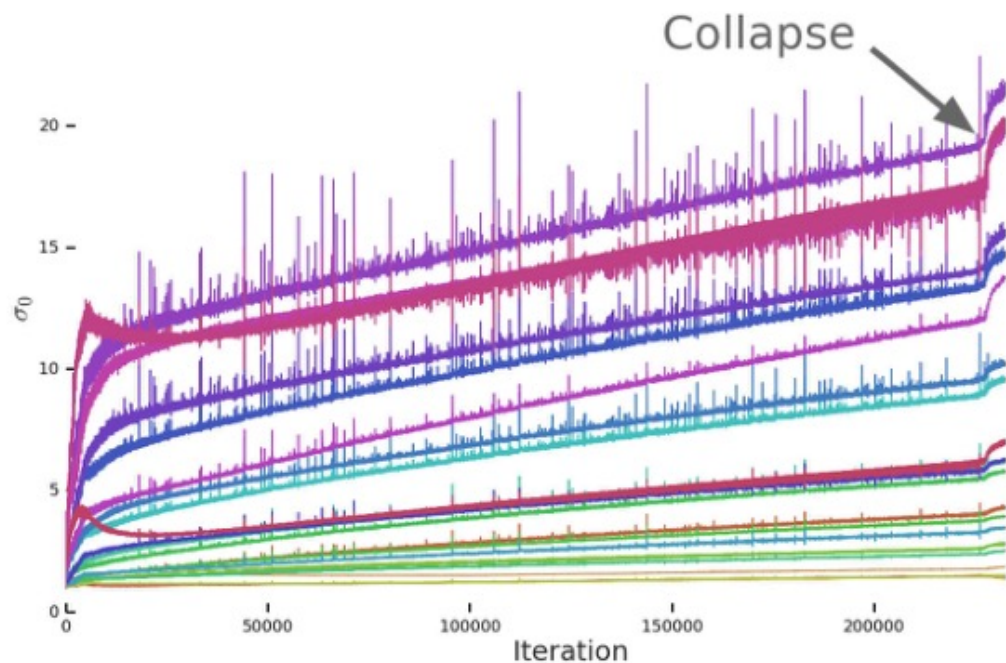
Analysis on Collapse

G: singular value 爆炸只是 collapse 會產生的現象



(a) G

D: singular value 非常雜亂，且非常容易 overfit 在 training set



(b) D

為什麼 GAN 的訓練會 collapse ?

監控每層權重的 singular value (預示 collapse 的 metric)

用 gradient penalty 來穩定訓練，但 generation quality 會下降非常多 (IS drop 45%)

結論：mode collapse 遲早會發生，在 collapse 前 early stop 是實務上最常見的做法

Experiments

IS: 只衡量像不像某一個類別的影像，不管多樣性
FID: 也會懲罰多樣性不足

without truncation

with truncation

Model	Res.	FID/IS	(min FID) / IS	FID / (valid IS)	FID / (max IS)
SN-GAN	128	27.62/36.80	N/A	N/A	N/A
SA-GAN	128	18.65/52.52	N/A	N/A	N/A
BigGAN	128	$8.7 \pm .6 / 98.8 \pm 3$	$7.7 \pm .2 / 126.5 \pm 0$	$9.6 \pm .4 / 166.3 \pm 1$	$25 \pm 2 / 206 \pm 2$
BigGAN	256	$8.7 \pm .1 / 142.3 \pm 2$	$7.7 \pm .1 / 178.0 \pm 5$	$9.3 \pm .3 / 233.1 \pm 1$	$25 \pm 5 / 291 \pm 4$
BigGAN	512	8.1/144.2	7.6/170.3	11.8/241.4	27.0/275
BigGAN-deep	128	$5.7 \pm .3 / 124.5 \pm 2$	$6.3 \pm .3 / 148.1 \pm 4$	$7.4 \pm .6 / 166.5 \pm 1$	$25 \pm 2 / 253 \pm 11$
BigGAN-deep	256	$6.9 \pm .2 / 171.4 \pm 2$	$7.0 \pm .1 / 202.6 \pm 2$	$8.1 \pm .1 / 232.5 \pm 2$	$27 \pm 8 / 317 \pm 6$
BigGAN-deep	512	7.5/152.8	7.7/181.4	11.5/241.5	39.7/298

Table 2: Evaluation of models at different resolutions. We report scores without truncation (Column 3), scores at the best FID (Column 4), scores at the IS of validation data (Column 5), and scores at the max IS (Column 6). Standard deviations are computed over at least three random initializations.

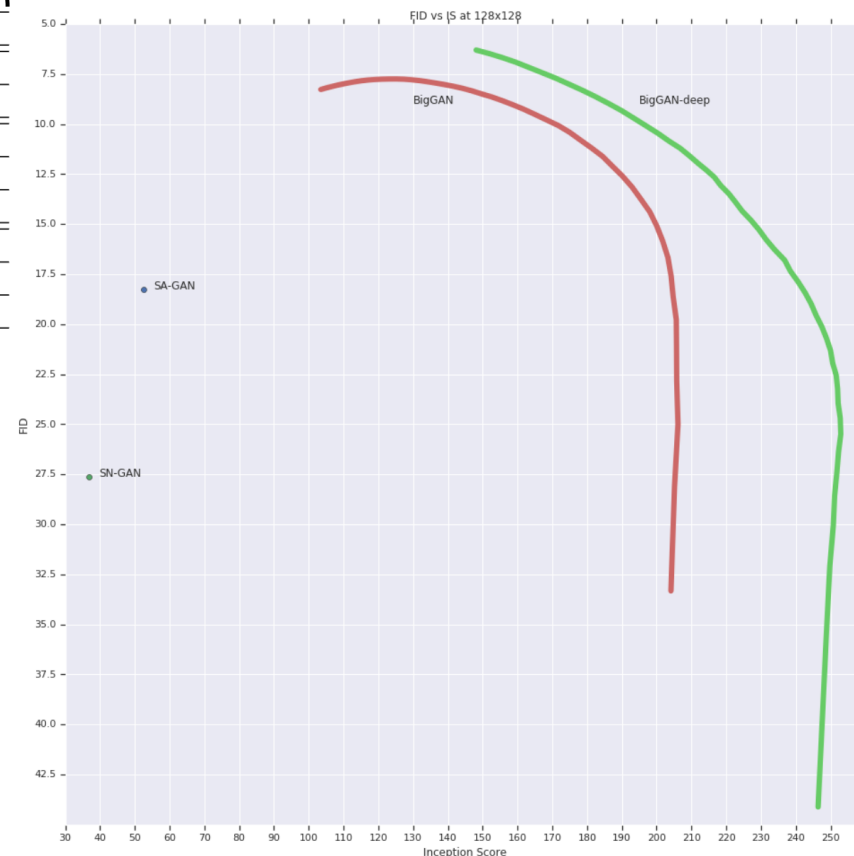


Figure 17: IS vs. FID at 128×128 . Scores are averaged across three random seeds.

Experiments

IS: 只衡量像不像某一個類別的影像，不管多樣性
FID: 也會懲罰多樣性不足

without truncation

with truncation

Model	Res.	FID/IS	(min FID) / IS	FID / (valid IS)	FID / (max IS)
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BigGAN-deep	256	$6.9 \pm .2 / 171.4 \pm 2$	$7.0 \pm .1 / 202.6 \pm 2$	$8.1 \pm .1 / 232.5 \pm 2$	$27 \pm 8 / 317 \pm 6$
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Table 2: Evaluation of models at different resolutions. We report scores without truncation (Column 3), scores at the best FID (Column 4), scores at the IS of validation data (Column 5), and scores at the max IS (Column 6). Standard deviations are computed over at least three random initializations.

和 training 時一樣直接 sample
from standard Normal distribution

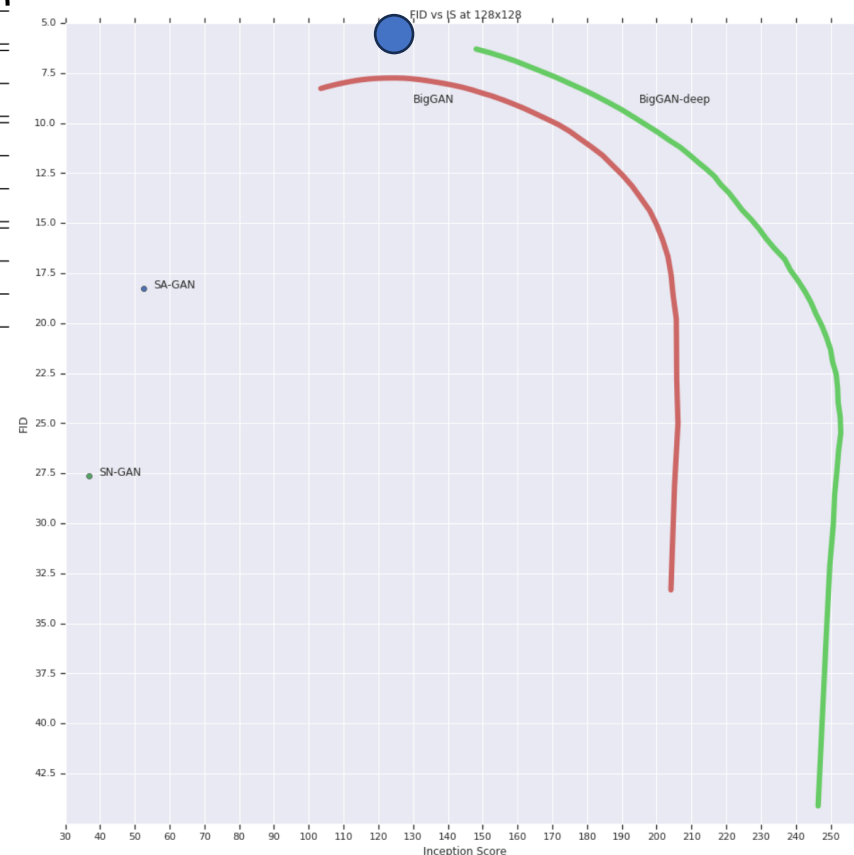


Figure 17: IS vs. FID at 128×128 . Scores are averaged across three random seeds.

Experiments

IS: 只衡量像不像某一個類別的影像，不管多樣性
FID: 也會懲罰多樣性不足

without truncation

with truncation

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掃過所有 truncation threshold，取
FID 最低（最好）的那一點（通常
也會是 IS 最差的一點）

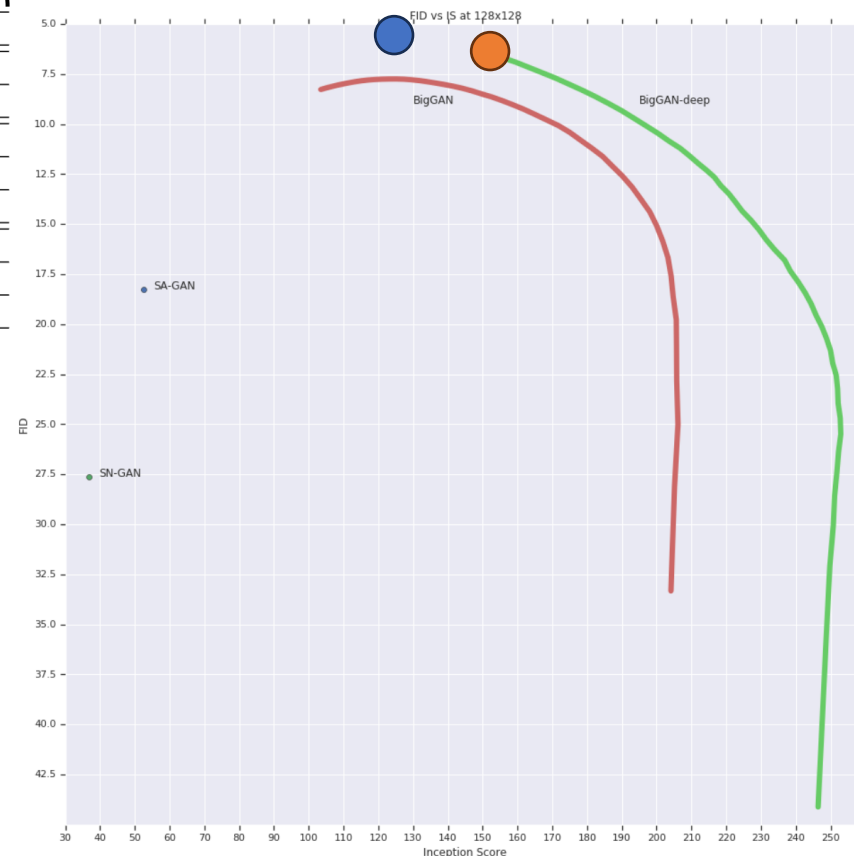


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先算真實 ImageNet validation set 在的 IS，把 threshold 調到模型的 IS 剛好等於這個值，這時候的 objectness 與真實的影像一樣好，看看在這條件下的多樣性能多好（這是作者覺得最有意義的一個 metric）

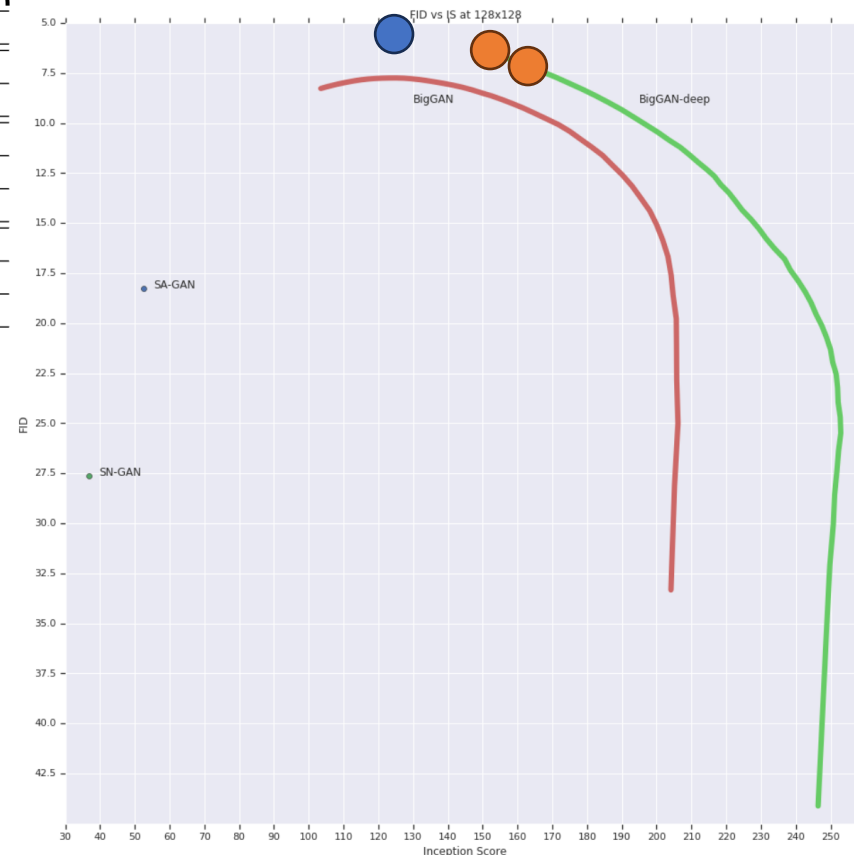


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把 threshold 壓到最小，這時候 IS 會最高，展示把 objectness 品質推到極致會犧牲多少多樣性

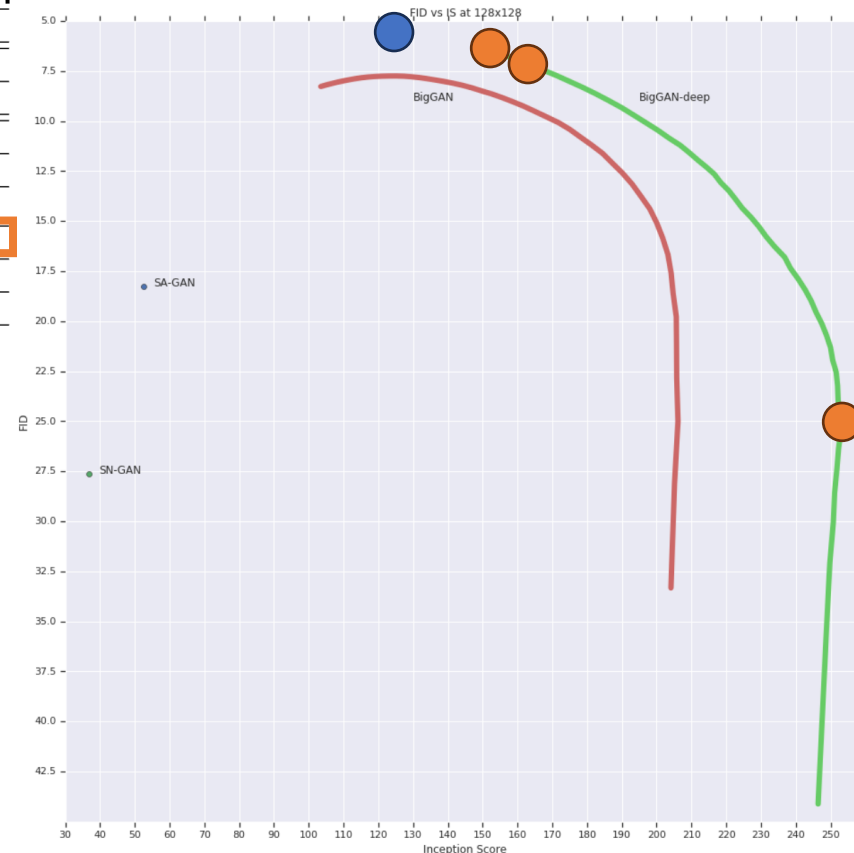


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BigGAN-deep 相比於 BigGAN 的層數更深四倍，且使用 **bottleneck** ResBlock，在所有解析度與指標都更好

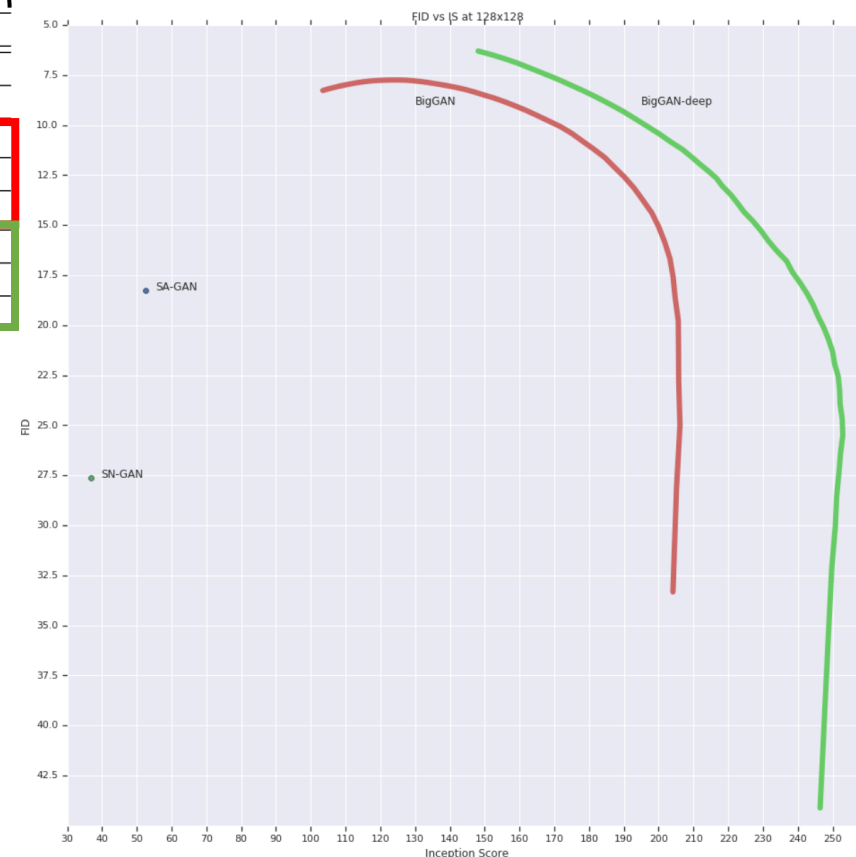


Figure 17: IS vs. FID at 128×128. Scores are averaged across three random seeds.

Experiments: 訓練在更多的資料 (JFT-300M)

同樣的技巧仍然有效，而且訓練不再 collapse



(a) 128×128



(b) 256×256



(c) 512×512



(d)

新的失敗模式 class leakage (狗臉長在網球上)，JFT-300M 有 8,500 類

Limitations

- Latent 沒有語意且無法控制（例如控制生成的頭的轉向或髮色）
- Truncation 這個 trick 很脆弱
- 無法利用 z 來產生隨機細節

能不能像 style transfer 那樣，把 latent 當作 style 餵進每一層來控制生成？



(b)

StyleGAN

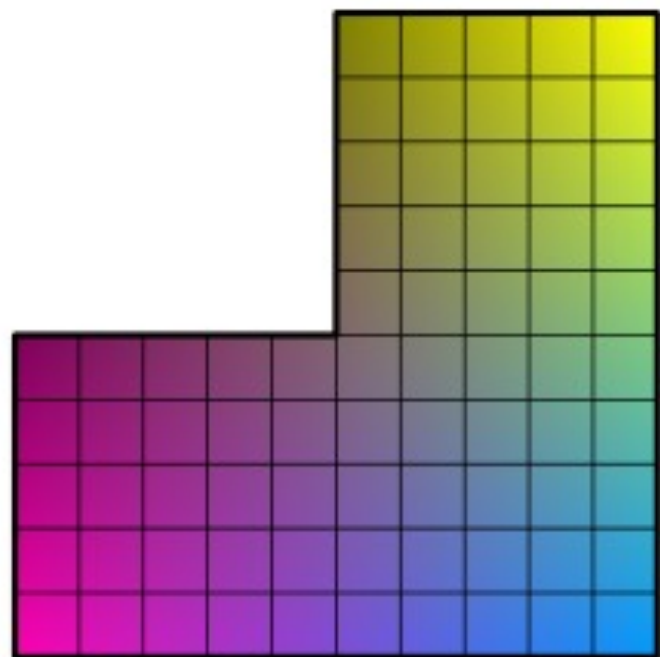
Karras et al., "A Style-Based Generator Architecture for GANs", CVPR 2019.

Style-based Generation

設計一個 generator 可以**自動**、**unsupervised** 地分離 high-level attributes (例如 pose、identity) 與 stochastic variation (雀斑、髮絲)，且能分尺度來控制



為什麼傳統 Generator 一定會 Entangled ?



(a) Distribution of features in training set

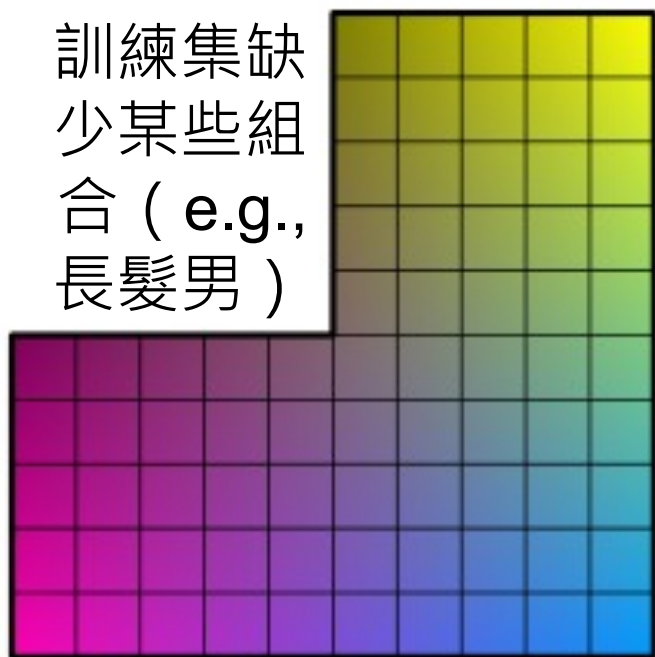
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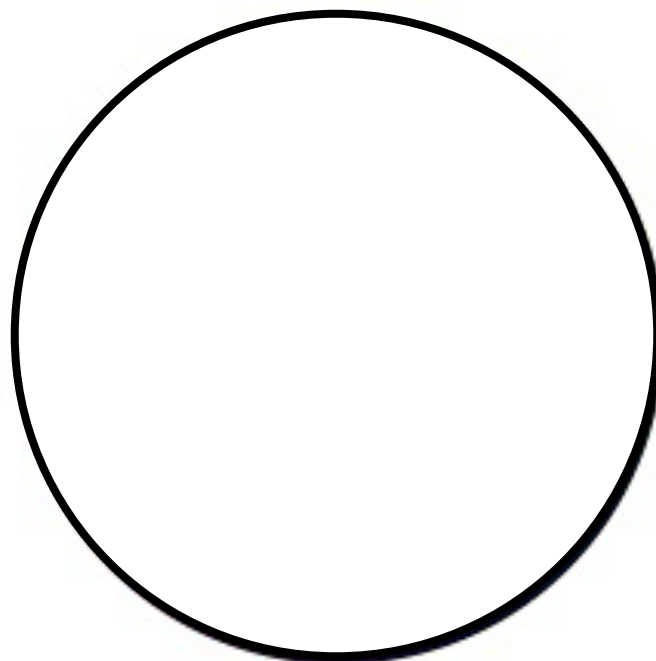
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訓練集缺少某些組合 (e.g., 長髮男)



(a) Distribution of features in training set



(b) Mapping from $\mathcal{Z}$ to features

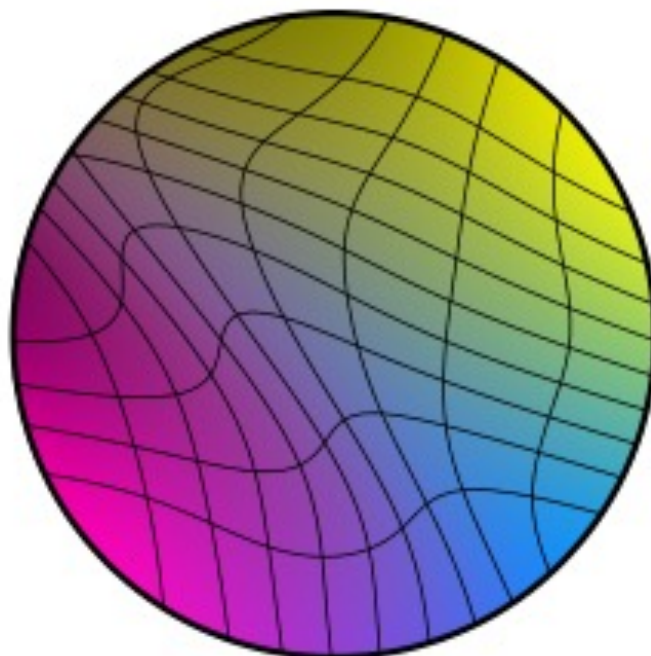
而 $\mathcal{Z}$ 是固定的 Gaussian distribution、取樣密度必須 mapping 對上資料密度

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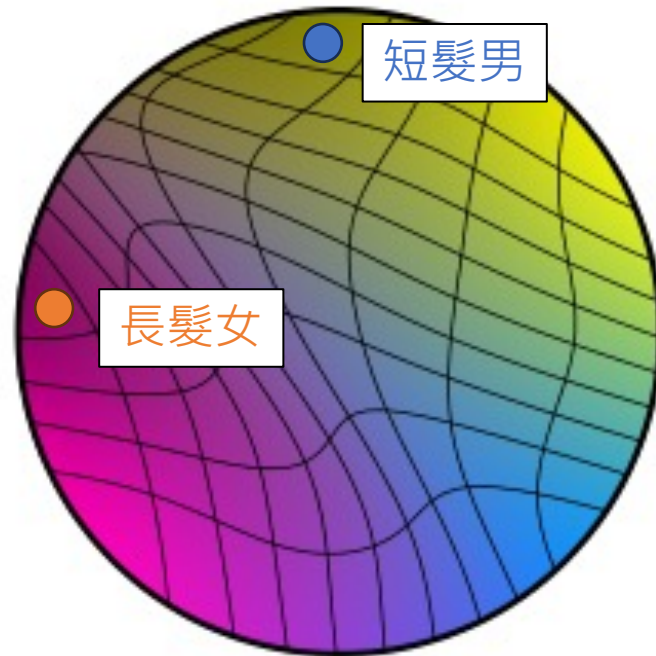
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因此 mapping 會被扭曲好讓缺失的組合在 Z 中消失，導致線性內插冒出無關的特徵

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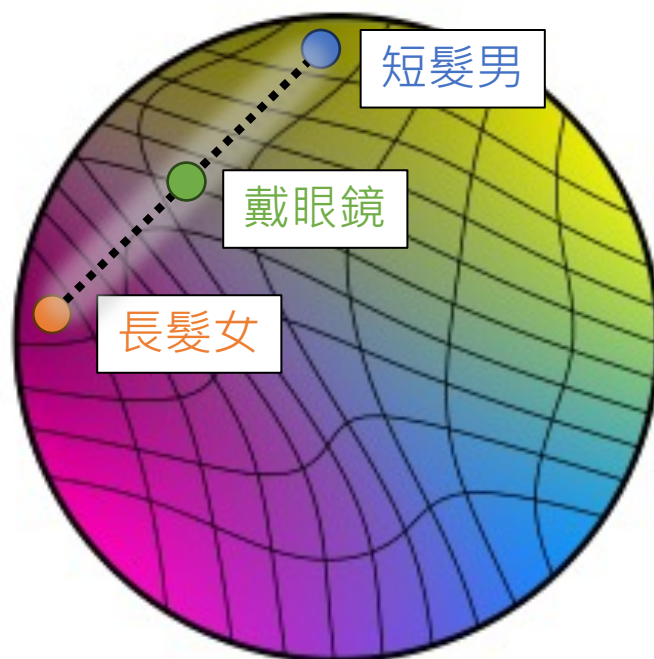
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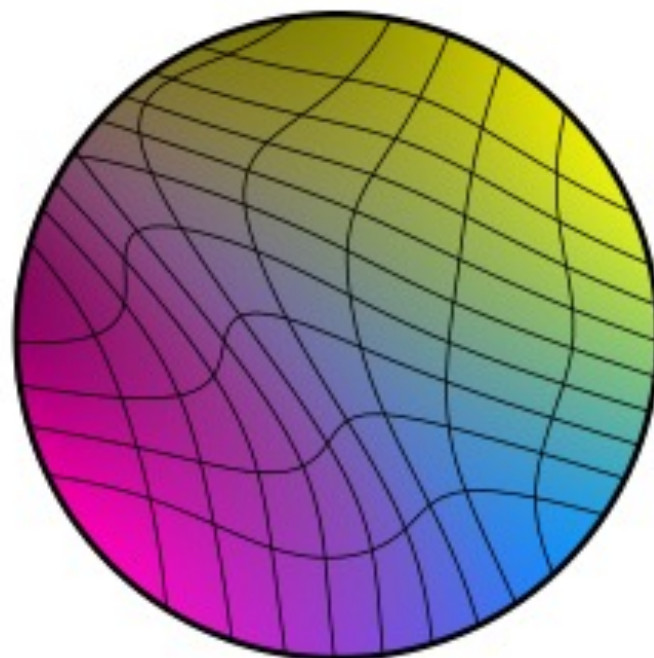
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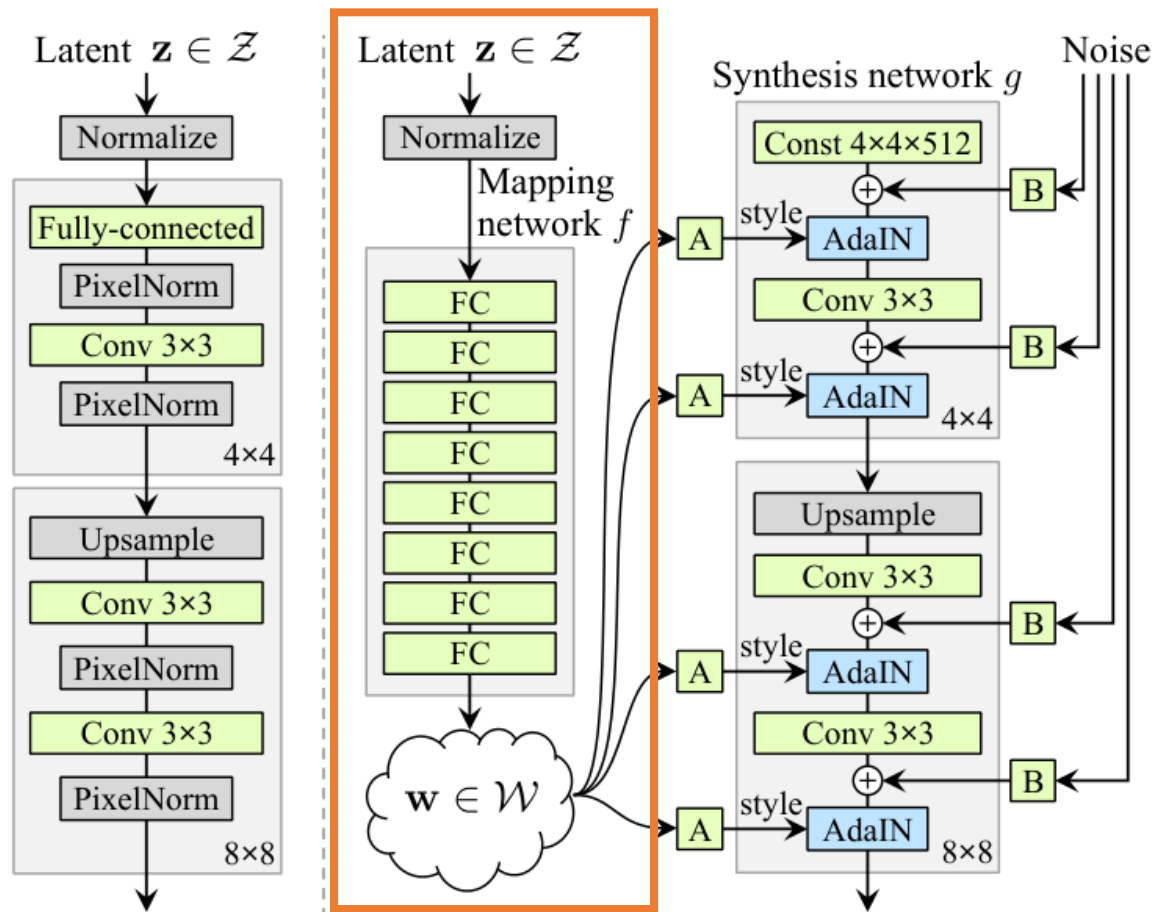
(b) Mapping from $\mathcal{Z}$ to features



(c) Mapping from $\mathcal{W}$ to features

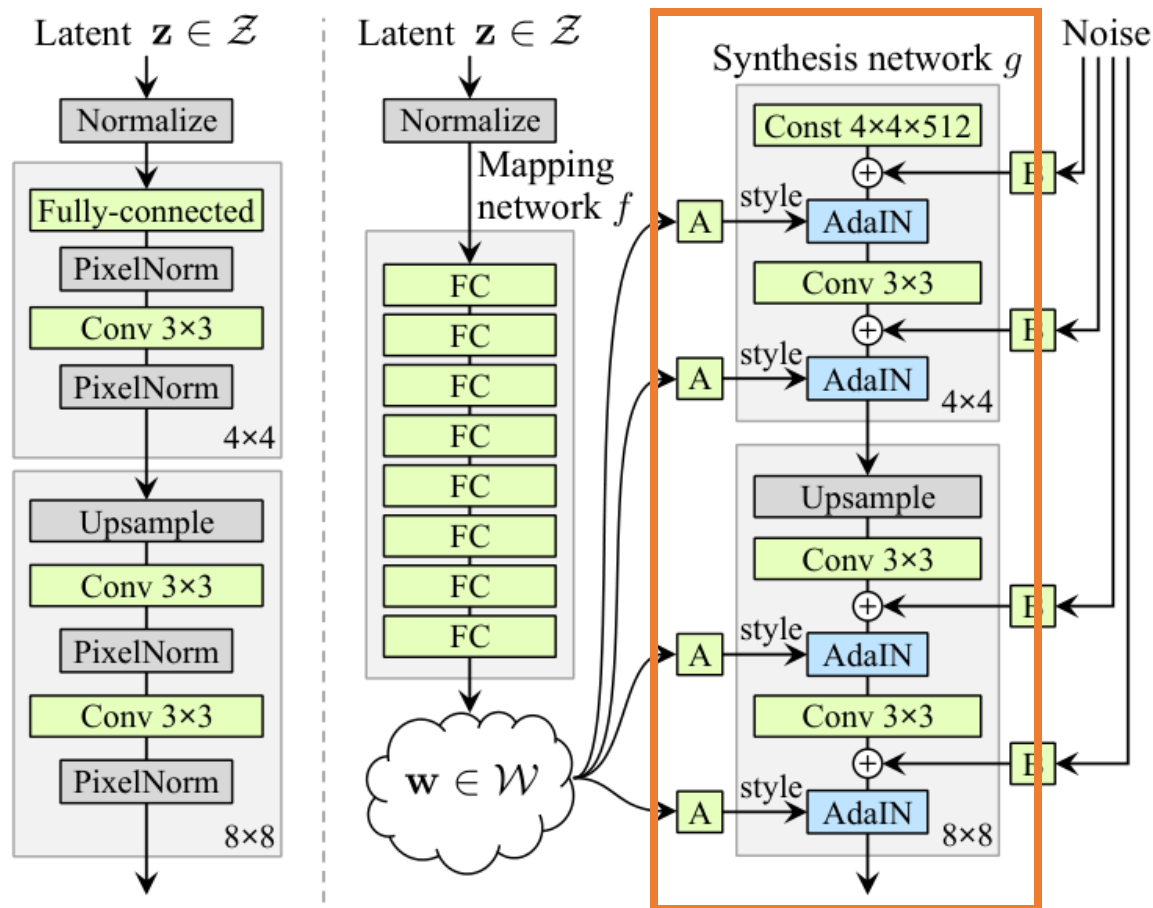
如果 latent 先經過一個 learnable 中繼 space $\mathcal{W}$ ($\mathcal{W}$ 不必符合任何固定分佈)，這個 learnable mapping network 可以把扭曲吸收掉，讓 $\mathcal{W} \rightarrow \text{mapping}$ 變成局部線性的

Method



- Mapping network $f: \mathbf{z} \rightarrow \mathbf{w}$
- 8 層 MLP
- $\mathbf{w}$ 經過可學的 affine transformation A 變成 style $\mathbf{y} = (\text{scale } y_s, \text{bias } y_b)$

Method

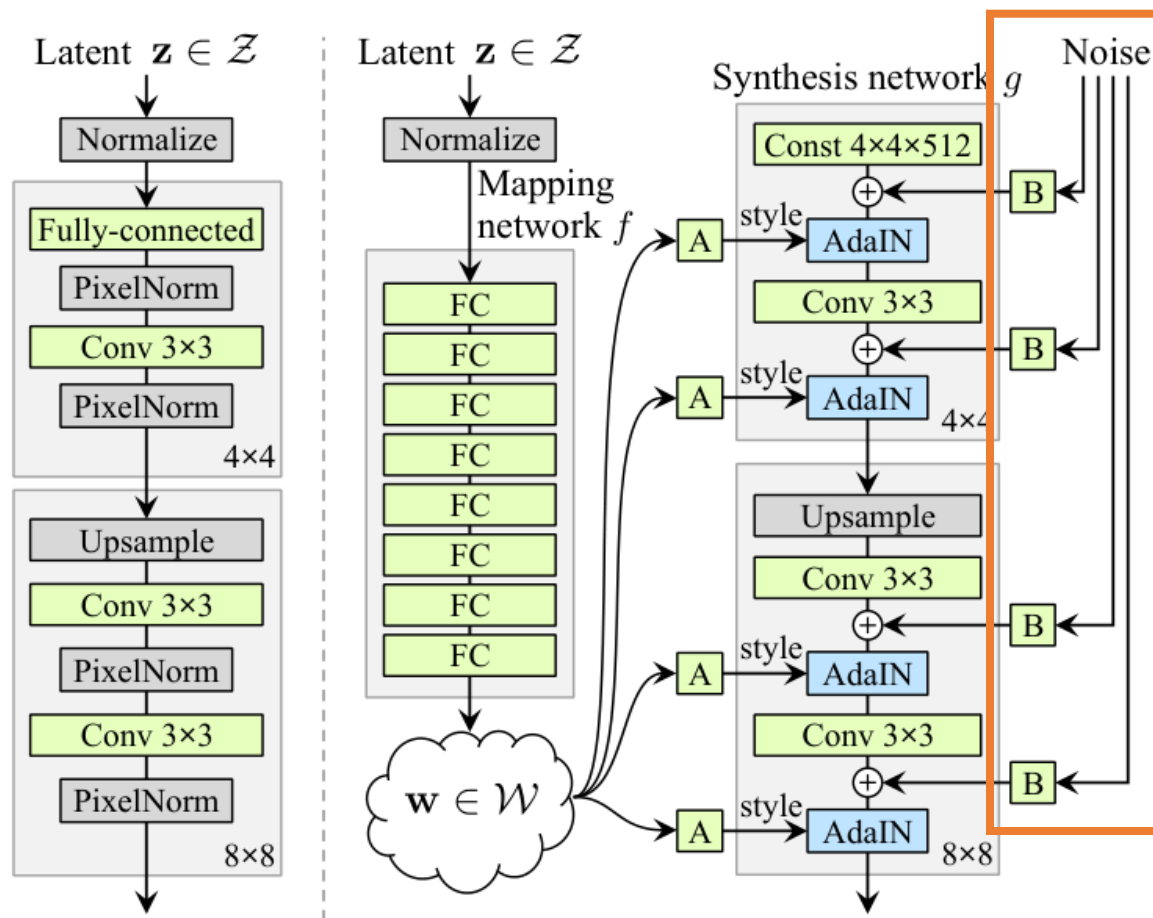


- Synthesis network 從一個 learned constant 4x4x512 出發
- 每個 conv 後接 **AdaIN**，用 style 決定該層每個 channel 的 scale y_s 與 bias y_b

$$\text{AdaIN}(x_i, y) = y_{s,i} \frac{x_i - \mu(x_i)}{\sigma(x_i)} + y_{b,i}$$

AdaIN: 先把每個 channel normalize 後再 apply style → 這個 layer 的統計量完全由 style 決定

Method

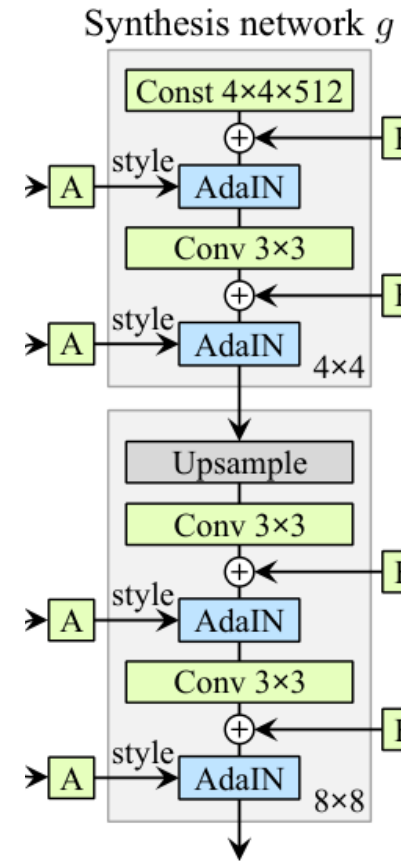


- Noise inputs : 每層加入一張獨立的 Gaussian noise image , 經過每個 channel 的 learnable scale B 後加到 conv 輸出

AdaIN 有局部性

- AdaIN 先把每個 channel normalize 到 zero mean / unit variance，再 apply style 的 scale 與 bias
- 這個 layer 的統計量完全由 style 決定、與前面無關
- 每個 style 只控制一個 conv，之後就被下一個 AdaIN 覆蓋

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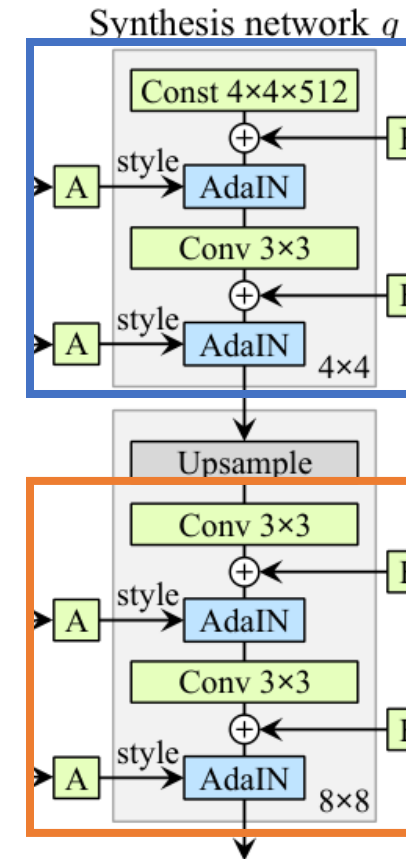


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這讓我們可以控制不同尺度的 style

$$\text{AdaIN}(x_i, y) = y_{s,i} \frac{x_i - \mu(x_i)}{\sigma(x_i)} + y_{b,i}$$



較 coarse 的 level: 例如 pose 臉型、髮型

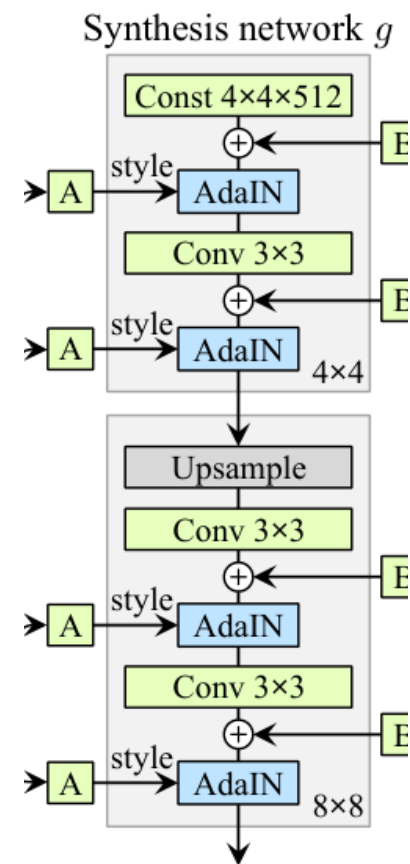
較 fine 的 level: 例如膚色、髮色、光線與細微結構

Mixing Regularization

如果訓練時每一層看到的 style 永遠來自同一個 w ，那 network 會學到把同一個屬性的資訊分散給好幾層之間互相補償（co-adaptation），這樣 style 就無法局部控制了

Mixing Regularization

- 訓練時一定比例（90%）的影像用兩個 latent w_1 、 w_2 ，在隨機的某個 layer 切換（這 layer 前都用 w_1 ，後面用 w_2 ）
- 這樣一來 network 就不能假設相鄰的 style 相關
- 這個機制類似 dropout 防止 neurons 共同適應，這邊是防止 cross layers 共同適應



AdaIN & Mixing Regularization

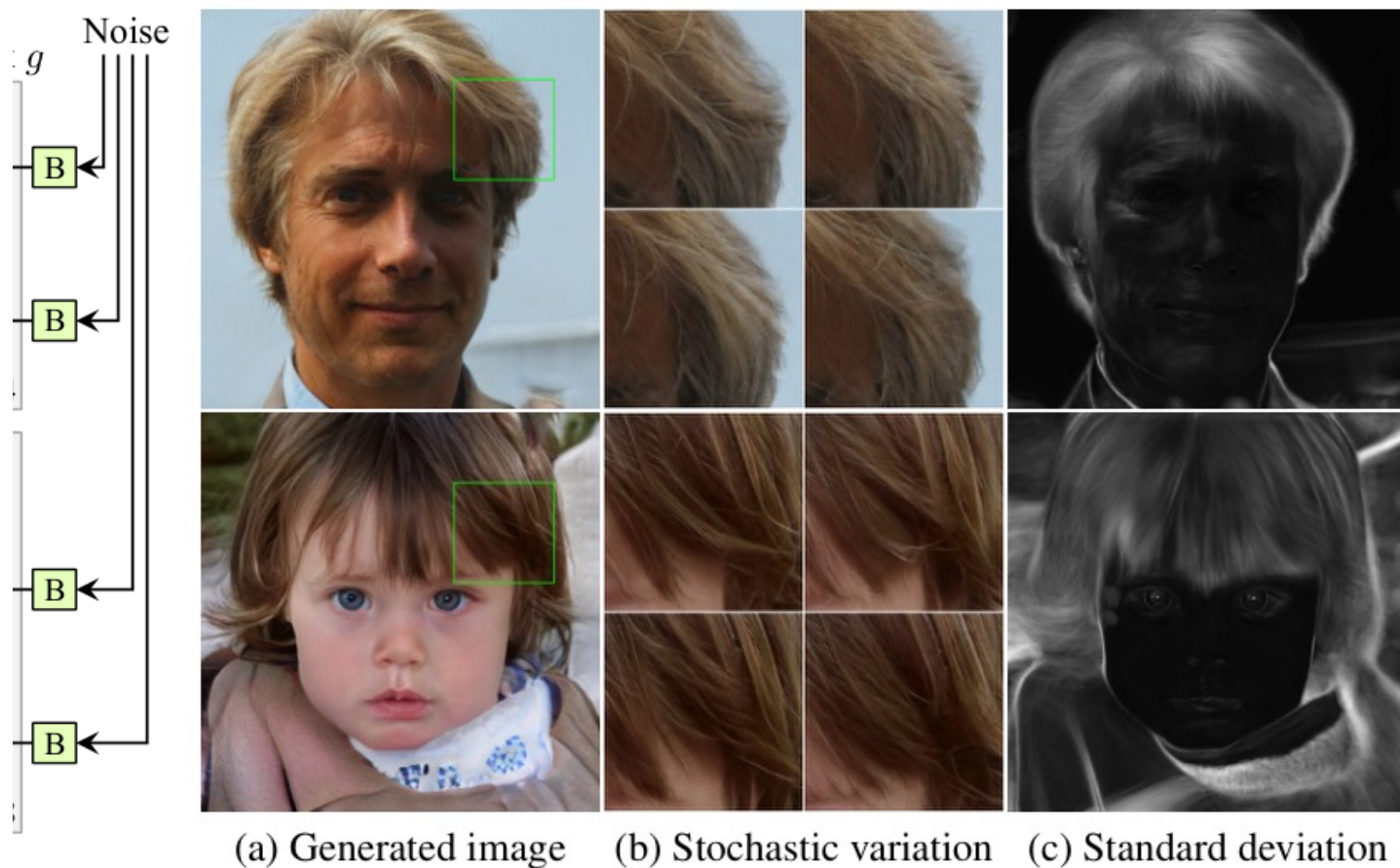
Coarse (4^2 - 8^2) :
pose、臉型、髮型、眼鏡

Middle (16^2 - 32^2) :
五官細節、眼睛開閉

Fine (64^2 - 1024^2) :
膚色、髮色、光線與細微結構

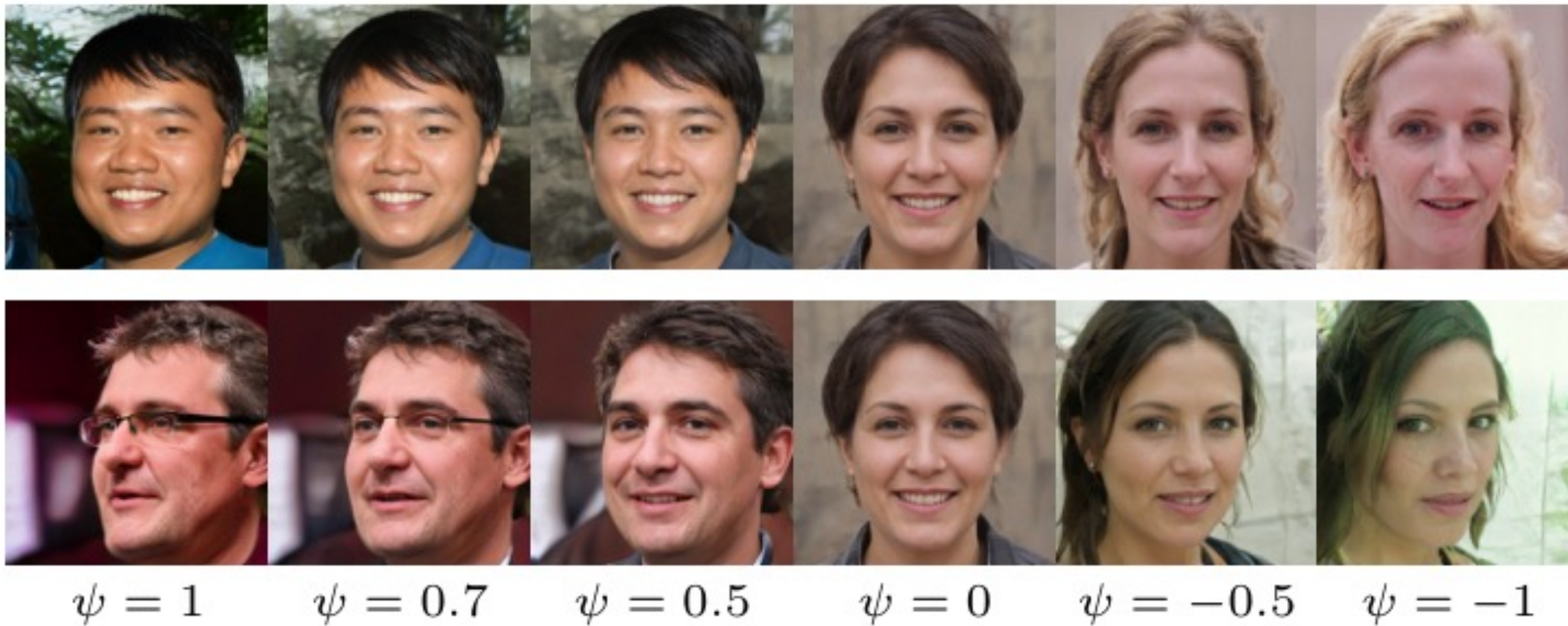


每層獨立 Noise 可以控制不同尺度的細節



- Noise 只會改髮絲、鬍渣、背景等細節，identity 與 pose 完全不會被改變
- Coarse 層的 noise 控制大範圍捲髮、背景形
- Fine 層的 noise 控制細髮絲、毛孔

Truncation in W



低密度區資料比較稀疏，所以 generator 一定 train 不好，truncation 可以在 generation 時避開這些區域
類似的手法在後面的 diffusion guidance (Classifier guidance 與 Classifier-free guidance) 中也有使用

Experiments

Method	CelebA-HQ	FFHQ
A Baseline Progressive GAN [30]	7.79	8.04
B + Tuning (incl. bilinear up/down)	6.11	5.25
C + Add mapping and styles	5.34	4.85
D + Remove traditional input	5.07	4.88
E + Add noise inputs	5.06	4.42
F + Mixing regularization	5.17	4.40

Table 1. Fréchet inception distance (FID) for various generator designs (lower is better). In this paper we calculate the FIDs using 50,000 images drawn randomly from the training set, and report the lowest distance encountered over the course of training.

Experiments

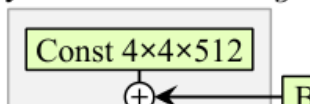
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style-based 比傳統 generator 好了將近 20%

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Synthesis network g



拿掉 input layer 對品質沒有影響

New Metrics & New Dataset FFHQ

- New metrics
 - Perceptual path length: 內插路徑上的 LPIPS 變化
 - linear separability: 屬性能否線性切分
- FFHQ
 - 70,000 張 1024^2 人臉，比 CelebA-HQ 更多樣， publicly available



From GAN to Diffusion

- GAN 的問題
 - Minimax optimization 不穩定、collapse 需要 early stop
 - Loss 不是一個簡單的 likelihood，很難用 loss 推斷訓練狀態
 - 要 scale 到開放領域的 text-to-image 很困難
- 但有些 GAN 的 components 在 diffusion 還是持續被採用
 - Stable Diffusion 的 **autoencoder** 就是使用 **patch-based** adversarial loss (pix2pix 的 PatchGAN)
 - Diffusion 的 **adaptive layer normalization** 是來自於 StyleGAN 的 **AdaIN**
 - Generation 十可以 trade-off between fidelity 與 diversity，在 GAN 是用 **truncation**，在 diffusion 是用 classifier-free guidance (**CFG**) 的 scale